

EEG-based Emotion Recognition Using Domain Adaptation Network

Yi-Ming Jin, Yu-Dong Luo, Wei-Long Zheng, Bao-Liang Lu*

Center for Brain-Like Computing and Machine Intelligence

Department of Computer Science and Engineering

Key Lab. of Shanghai Education Commission for Intelligent Interaction and Cognitive Eng.

Brain Science and Technology Research Center

Shanghai Jiao Tong University

800 Dong Chuan Road, Shanghai 200240, China

Email: {jinyiming, miyunluo, weilong, bllu}@sjtu.edu.cn

Abstract—This paper explores a fundamental problem of eliminating the differences between source subject and target subject in EEG-based emotion recognition. The major limitation of traditional classification methods is that the lack of domain adaptation and subspace alignment will degrade the performance of cross-subject emotion recognition. To address this problem, we adopt Domain Adaptation Network (DAN) for knowledge transfer, which maintains both feature discriminativeness and domain-invariance during training stage. A feed-forward neural network is constructed by augmenting a few standard layers and a gradient reversal layer. Compared with five traditional methods, DAN outperforms its counterparts and achieves the mean accuracy of 79.19%. Moreover, a visualization of the features learned by DAN is represented in this paper, which intuitively describes the transfer virtue of domain adaptation network.

Keywords—EEG; emotion recognition; domain adaptation network; transfer learning;

I. INTRODUCTION

Emotion is normal but essential to humans. It not only contributes to interaction between humans, but also plays a critical role in rational and intelligent behavior. With the development of wearable EEG devices and affective Brain-Computer Interface (aBCI) [1], researchers begin to use EEG for emotion recognition.

A variety of classifiers have been constructed for EEG-based emotion recognition, such as k -nearest neighbor (kNN), logistic regression (LR), support vector machine (SVM), and deep belief network (DBN) [2]. However, these classifiers are not suitable for cross subject emotion recognition due to the variability between subjects and sessions.

Domain adaptation (DA) scenario is investigated to address this problem [3]. Typically, training samples with class labels are defined as source domain and testing samples with different distributions are called target domain [4]. In this work, we denote $X \in \mathcal{X}$ as an input of (X, y) from the EEG recording, where $y \in \mathcal{Y}$ is the corresponding emotion label. Let C be the number of

channels and d be the number of time series samples, then $\mathcal{X} = \mathcal{R}^{C \times d}$ in this circumstance. According to [4], $\mathcal{D} = \{\mathcal{X}, P(X)\}$ is a domain, where $P(X)$ is the marginal probability distribution of X . In our case, this domain contains given subjects from which we recorded the EEG signals. The source and target domains in this paper share the same feature space, $\mathcal{X}_S = \mathcal{X}_T$, but differ in marginal probability distributions, which means $P(X_S) \neq P(X_T)$. The key assumption in most domain adaptation methods is that the conditional probability distributions are the same, i.e. $P(Y_S|X_S) = P(Y_T|X_T)$ [5].

A crucial issue for subject transfer is how to reduce the discrepancy between source domain and target domain. Several approaches are proposed to minimize the variability of EEG signals among subjects. Zheng *et al.* [6] applied transfer component analysis (TCA) [7] and kernel principle component analysis (KPCA) [8] for feature selection and reduction. Besides, Zheng *et al.* [5] adopted transductive parameter transfer (TPT) [9] method to personalize EEG-based affective models, which achieves a significant improvement in subject transfer.

Inspired by the transferability of deep neural networks, in this paper we propose a EEG-based emotion recognition method by adopting domain adaptation network (DAN) which learns domain-invariant features by backpropagation. We compare the performance of our work with three state-of-the-art approaches, TCA, KPCA and TPT.

II. METHOD

A. Domain Adaptation Network

We use a domain adaptation network (DAN) to achieve transfer learning. This framework is proposed by Yaroslav Ganin *et al.* [10] for image classification. This architecture consists of three parts as shown in Figure 1. We denote $\mathcal{S}(x)$ and $\mathcal{T}(x)$ as the marginal distributions of source and target domain, respectively. Given the training dataset $\{x_1, x_2, \dots, x_N\}$ concatenated from both source and target data, we introduce a binary variable $d_i \in \{0, 1\}$ serving as the domain label for the i -th sample. It points out which domain distribution x_i comes from, i.e. $x_i \sim \mathcal{S}(x)$ if $d_i = 0$ and $x_i \sim \mathcal{T}(x)$ if $d_i = 1$.

For each input data x , a feature vector $f \in \mathbf{R}^D$ is firstly learned by a feature extractor M_f , i.e. $f = M_f(x, \theta_f)$, where θ_f denotes parameters of all layers in this mapping.

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基于脑电的情绪识别——使用领域自适应网络

金一鸣, 罗宇东, 郑伟龙, 陆宝良
脑计算与机器智能中心 计算机科学与工程学院 上海教育委员会智能交互与认知工程重点实验室 脑科学与技术研究中心 上海交通大学 800 东川路, 上海 200240, 中国 邮箱: {jinyiming, miyunluo, weilong, bllu}@sjtu.edu.cn

摘要—本文探讨了一种消除脑电图 (EEG) 情绪识别中源主体与目标主体差异的基本问题。传统分类方法的主要局限性在于缺乏领域适应和子空间对齐, 这将降低跨主体情绪识别的性能。为解决这一问题, 我们采用领域适应网络 (DAN) 进行知识迁移, 该网络在训练阶段同时保持特征判别性和领域不变性。通过增加几层标准层和一个梯度反转层, 构建了一个前馈神经网络。与五种传统方法相比, DAN 表现更优, 实现了 79.19% 的平均准确率。此外, 本文还展示了 DAN 学习到的特征的可视化结果, 直观地描述了领域适应网络的迁移优势。
关键词—脑电图; 情绪识别; 领域适应网络; 迁移学习;

I. 我 情绪对人类来说是正常的, 但也是必不可少的。它不仅促进了人类之间的互动, 还在理性和智能行为中发挥着关键作用。随着可穿戴脑电图 (EEG) 设备和情感脑机接口 (aBCI) [1] 的发展, 研究人员开始使用 EEG 进行情绪识别。为了进行基于 EEG 的情绪识别, 已经构建了多种分类器, 如 k 近邻 (kNN)、逻辑回归 (LR)、支持向量机 (SVM) 和深度信念网络 (DBN) [2]。然而, 由于受试者之间的差异和不同会话中的变化, 这些分类器并不适用于跨受试者的情绪识别。为了解决这个问题, 研究了领域自适应 (DA) 场景[3]。通常, 带有类别标签的训练样本被定义为源域, 而具有不同分布的测试样本被称为目标域[4]。在本工作中, 我们用 $X \in \mathcal{X}$ 表示来自 EEG 记录的 (X, y) 的输入, 其中 $y \in \mathcal{Y}$ 是相应的情绪标签。设 $\mathcal{C}$ 为

如果 channels 和 d 是时间序列样本的数量, 那么 $X = R$ 在这种情况下。根据[4], $D = \{X, P(X)\}$ 是一个域, 其中 $P(X)$ 是 X 的边缘概率分布。在我们的案例中, 这个域包含我们从中记录脑电图信号的给定主题。本文中的源域和目标域共享相同特征空间, $X = X$, 但在边缘概率分布上有所不同, 这意味着 $P(X) \neq P(X)$ 。大多数域适应方法中的关键假设是条件概率分布相同, 即 $P(Y|X) = P(Y|X)$ [5]。

对于主题迁移的一个关键问题是如何减少源域和目标域之间的差异。已经提出了几种方法来最小化受试者之间脑电图 (EEG) 信号的变异性。Zheng 等人[6]应用了迁移成分分析 (TCA) [7] 和核主成分分析 (KPCA) [8] 进行特征选择和降维。此外, Zheng 等人[5]采用了有监督参数迁移 (TPT) [9] 方法来个性化基于 EEG 的情感模型, 这在主题迁移方面取得了显著改进。

受深度神经网络可迁移性的启发, 在本文中我们提出了一种基于 EEG 的情绪识别方法, 通过采用域适应网络 (DAN) 来通过反向传播学习域不变特征。我们将我们的工作与三种最先进的方法 (TCA、KPCA 和 TPT) 的性能进行了比较。

II. M

A. 域适应网络

我们使用领域自适应网络 (DAN) 来实现迁移学习。该框架由 Yaroslav Ganin 等人[10] 为图像分类提出。该架构由图 1 所示的三部分组成。我们分别将 $S(x)$ 和 $T(x)$ 表示为源域和目标域的边缘分布。给定由源数据和目标数据连接而成的训练数据集 $\{x, x, \dots, x\}$, 我们引入一个二元变量 $d \in \{0, 1\}$ 作为第 i 个样本的领域标签。它指出 x 来自哪个领域分布, 即如果 $d =$

0 并且当 $d=1$ 时, $x \sim T(x)$
对于每个输入数据 x , 特征提取器 M 首先学习一个特征向量 $f \in \mathbb{R}$, 即 $f = M(x, \theta)$, 其中 θ 表示该映射中所有层的参数。

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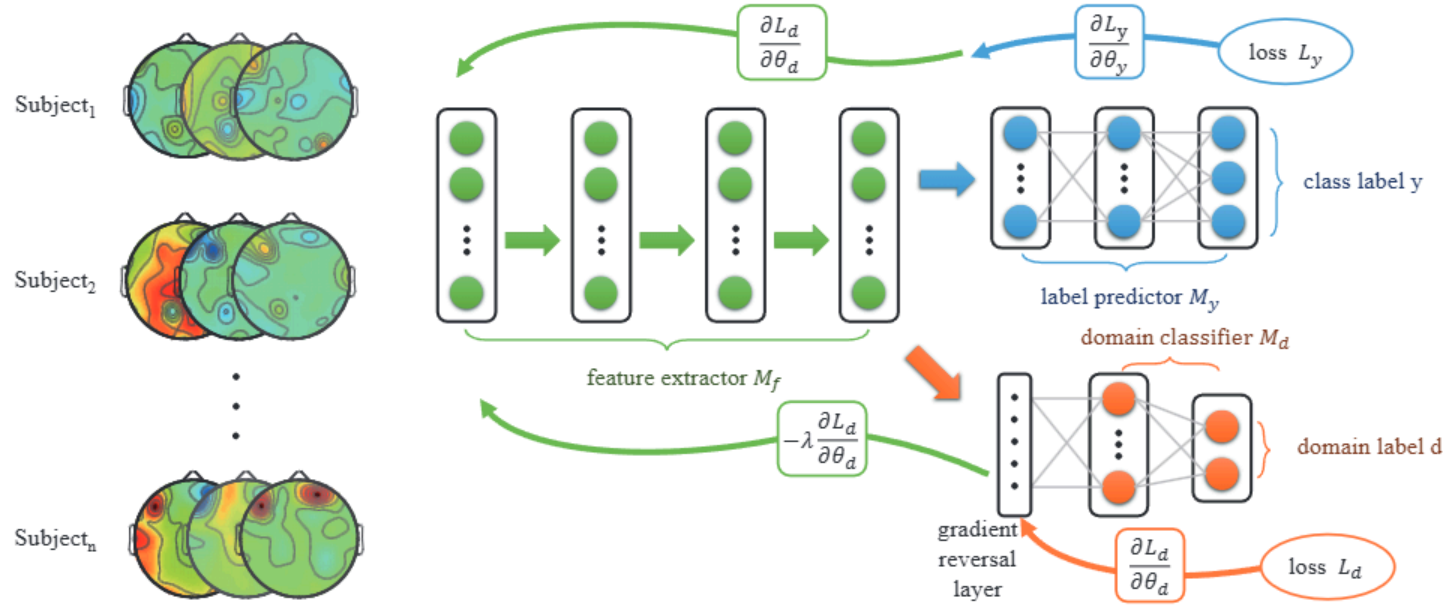


Figure 1. Domain Adaptation Network contains a feature extractor (green), a label predictor (blue), a domain classifier (red) and a gradient reversal layer (black). Feature extractor and label predictor together form a traditional feed-forward neural network. These two parts can be used to predict the label. Domain adaptation is achieved by feature extractor and domain classifier by backpropagation. The gradient conveyed from domain classifier to feature extractor is multiplied by a negative parameter in gradient reversal layer. In this way, traditional gradient descent method can be applied. Label predictor ensures the data discriminativeness and domain classifier maintains the domain-invariance.

Then the feature vector f is mapped by two parallel networks. The label predictor M_y maps f with parameters θ_y and the domain classifier M_d maps f with parameters θ_d .

The label predictor M_y is used to predict labels of the source domain. In order to get a good prediction performance, we intend to minimize the loss of label predictor on the source domain. Consequently, both parameters θ_f and θ_y are adjusted during the training time aiming to minimize the experiential loss for source domain samples.

The domain classifier M_d indicates the domain label d for each input data. The network learns domain-invariant feature f from input data, which means the distributions $\mathcal{S}(f) = \{M_f(x, \theta_f) | x \sim \mathcal{S}(x)\}$ and $\mathcal{T}(f) = \{M_f(x, \theta_f) | x \sim \mathcal{T}(x)\}$ should be similar. According to *covariance shift* assumption [11], the features are transferable and the label prediction accuracy on target domain matches that on source domain in this case. To tackle this problem, the loss of domain classifier M_d is used to measure the domain discrepancy.

The loss function of the architecture can be computed as

$$\begin{aligned} E(\theta_f, \theta_y, \theta_d) &= \sum_{\substack{i=1 \dots N \\ d_i=0}} L_y(M_y(M_f(x_i; \theta_f); \theta_y), y_i) - \\ &\quad \lambda \sum_{i=1 \dots N} L_d(M_d(M_f(x_i; \theta_f); \theta_d), y_i) \\ &= \sum_{\substack{i=1 \dots N \\ d_i=0}} L_y^i(\theta_f, \theta_y) - \lambda \sum_{i=1 \dots N} L_d^i(\theta_f, \theta_d) \end{aligned} \quad (1)$$

where $L_y(\cdot, \cdot)$ is the loss of class label prediction, $L_d(\cdot, \cdot)$ represents the loss of domain label classification, L_y^i and L_d^i denote the corresponding loss functions of i -th training example, and λ is the trade-off parameter that balances the two objectives.

During learning process, a method similar to the s-

tandard stochastic gradient descent (SGD) is adopted to search the saddle point: $\theta_f \leftarrow \theta_f - \mu(\frac{\partial L_y^i}{\partial \theta_f} - \lambda \frac{\partial L_d^i}{\partial \theta_f})$, $\theta_y \leftarrow \theta_y - \mu \frac{\partial L_y^i}{\partial \theta_y}$, $\theta_d \leftarrow \theta_d - \mu \frac{\partial L_d^i}{\partial \theta_d}$, where μ denotes the learning rate.

The updating process is similar to SGD except for parameters θ_f which combine gradients from label predictor and domain classifier. However, directly implementing it using SGD is difficult. Ganin *et al.* [10] adds a gradient reversal layer (GRL) between feature extractor and domain classifier. During forward propagation, GRL acts as an identity transformation which does not change the input parameters. While in backpropagation, GRL multiplies $-\lambda$ to the gradient from subsequent layer and passes the result to the preceding layer. More precisely, the GRL can be regarded as a pseudo-function $R_\lambda: R_\lambda = x, \frac{dR_\lambda}{dx} = -\lambda I$, where I is an identity matrix. With the help of such transformation, the loss function E can be further written as:

$$\begin{aligned} \tilde{E}(\theta_f, \theta_y, \theta_d) &= \sum_{\substack{i=1 \dots N \\ d_i=0}} L_y(M_y(M_f(x_i; \theta_f); \theta_y), y_i) + \\ &\quad \sum_{i=1 \dots N} L_d(M_d(R_\lambda(M_f(x_i; \theta_f))); \theta_d, y_i) \end{aligned} \quad (2)$$

After optimization, the feature extractor $M_f(\cdot, \cdot)$ and label predictor $M_y(\cdot, \cdot)$ can be used in target domain to predict the label.

III. EXPERIMENT SETUP

A. EEG Dataset

In this paper, we evaluate the performance of domain adaptation network (DAN) using a publicly available EEG dataset for emotion recognition called SJTU Emotion EEG dataset (SEED)¹ downloaded from the project website. Different from other existing EEG dataset, each participant

¹<http://bcmi.sjtu.edu.cn/~seed/>

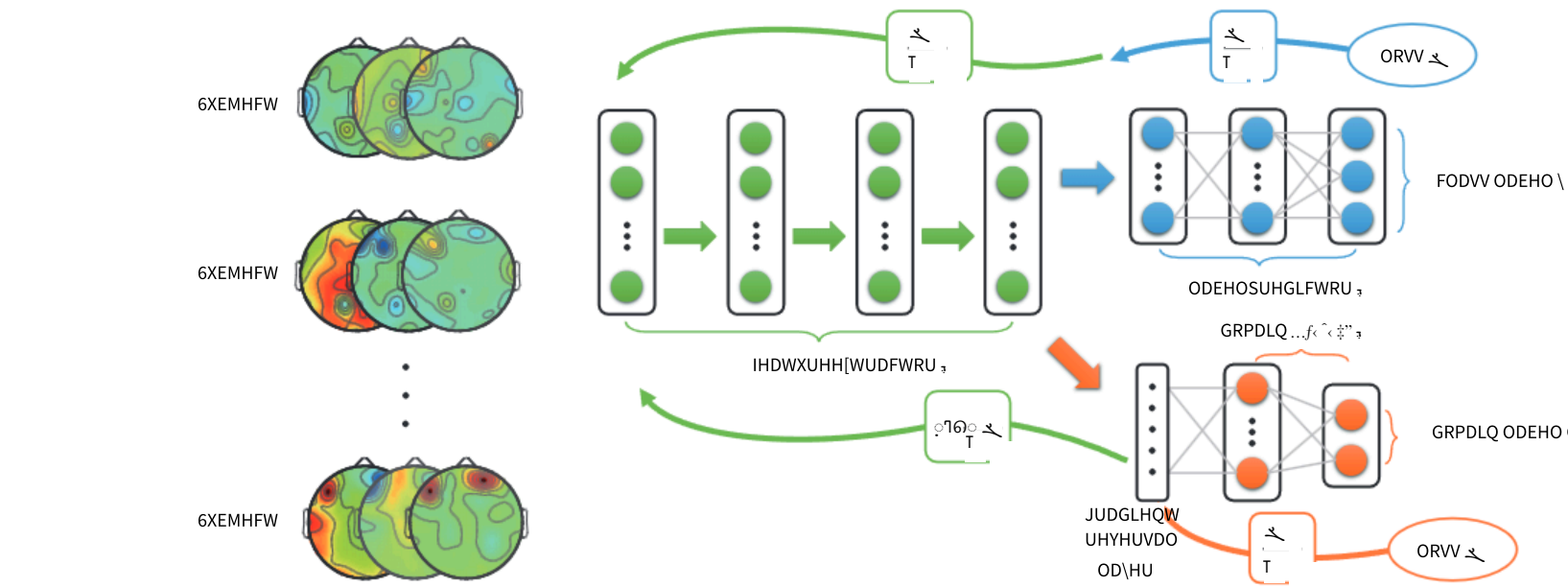


图 1. 域适应网络包含一个特征提取器（绿色）、一个标签预测器（蓝色）、一个域分类器（红色）和一个梯度反转层（黑色）。特征提取器和标签预测器共同构成一个传统的前馈神经网络。这两部分可以用来预测标签。通过特征提取器和域分类器的反向传播实现域适应。从域分类器传递到特征提取器的梯度在梯度反转层中乘以一个负参数。这样，传统的梯度下降方法可以应用。

标签预测器确保数据的区分性，域分类器保持域不变性。

然后特征向量 f 被两个并行网络映射。标签预测器 M 使用参数 θ 映射 f ，域分类器 M 使用参数 θ 映射 f 。

标签预测器 M_{is} 用于预测源域的标签。为了获得良好的预测性能，我们打算在源域上最小化标签预测器的损失。因此，在训练期间调整参数 θ 和 θ ，旨在最小化源域样本的经验损失。

域分类器 M_{ind} 表示每个输入数据的域标签 d 。网络从输入数据中学习域不变特征 f ，这意味着分布 $S(f) = \{M(x, \theta) \mid x \sim S(x)\}$ 和 $T(f) = \{M(x, \theta) \mid x \sim T(x)\}$ 应该是相似的。根据协方差偏移假设[11]，特征是可以迁移的，在这种情况下，目标域上的标签预测精度与源域上的精度相同。为了解决这个问题，域分类器 M_{is} 的损失被用来衡量域差异。

该架构的损失函数可以计算为

$$E(\theta, \theta, \theta) = \sum_{i=1 \dots N} L(M(M(x; \theta); \theta), y) - \lambda \sum_{d=0} L(M(M(x; \theta); \theta), y) = \sum_{i=1 \dots N} L(\theta, \theta) - \lambda \sum_{i=1 \dots N} L(\theta, \theta)$$

(1) 其中 $L(\cdot, \cdot)$ 是类别标签预测的损失， $L(\cdot, \cdot)$ 表示领域标签分类的损失， L_{and} 和 L 分别表示第 i 个训练样本的对应损失函数， λ 是平衡两个目标的权衡参数。

标准随机梯度下降（SGD）被用来搜索鞍点： $\theta \leftarrow \theta - \mu \left(\frac{\partial L}{\partial \theta} - \lambda \frac{\partial L}{\partial \theta} \right)$

$\theta \leftarrow \theta - \mu \frac{\partial L}{\partial \theta}, \theta_d \leftarrow \theta_d - \mu \frac{\partial L}{\partial \theta_d}$, 其中 μ 表示学习率。

更新过程与 SGD 类似，除了参数 θ 结合了标签预测器和域分类器的梯度。然而，直接使用 SGD 实现它很困难。Ganin 等人 [10] 在特征提取器和域分类器之间添加了一个梯度反转层

（GRL）。在前向传播过程中，GRL 充当恒等变换，不会改变输入参数。而在反向传播过程中，GRL 将后续层的梯度乘以 $-\lambda$ ，然后将结果传递到前一层。更精确地说，GRL 可以被视为一个伪函数 $R: R=x, dx=-\lambda I$ ，其中 I 是恒等矩阵。借助这种变换，损失函数 E 可以进一步写为：

$$\tilde{E}(\theta, \theta, \theta) = \sum_{i=1 \dots N} L(M(M(x; \theta); \theta), y) + \sum_{d=0} L(M(R(M(x; \theta)); \theta), y)$$

(2) 优化后，特征提取器 $M(\cdot, \cdot)$ 和标签预测器 $M(\cdot, \cdot)$ 可用于目标域以预测标签。

III. E S

A. 脑电图数据集

在本文中，我们使用一个公开可用的脑电图数据集来评估领域自适应网络（DAN）在情绪识别方面的性能，该数据集名为上海交通大学情绪脑电图数据集（SEED），从项目网站下载。与其他现有的脑电图数据集不同，每个参与者

在学习过程中，采用一种类似于 s-

1 <http://bcmi.sjtu.edu.cn/~seed/>

performs experiments three times at intervals of one week or longer and is asked to assess his or her real emotional reactions right after watching a movie clip. Only the data with explicit corresponding emotions are used for analysis. For more details of SEED, we recommend readers to refer to the literature [2].

B. Signal Preprocessing and Feature Extraction

For signal preprocessing, since raw EEG signals are often contaminated by electromyography (EMG) signals and electrooculogram (EOG) signals [12], a bandpass filter between 1 Hz and 75 Hz is used to filter out noise and artifacts. To reduce the data size, EEG signals are further down-sampled to 200 Hz.

For feature extraction, differential entropy (DE) features are extracted from raw data since DE outperforms power spectral density (PSD) [13]. DE features contain five frequency bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-14 Hz), beta (14-31 Hz), and gamma (31-50 Hz). Each frequency band contains the features of 62 EEG channels, thus the total dimensions of extracted EEG features are 310.

C. Parameter Details

All the evaluations are conducted using leave-one-subject-out cross validation. Each time, one subject is selected as target domain out of 15 subjects and the rest 14 remain as source domain. Classification accuracy is evaluated on the target domain.

Baselines. We adopt Support Vector Machine (SVM) as one of our baseline. We use a linear kernel and search the parameter C from -10 to 10 with step of 1. To compare with DAN, we also adopt traditional multilayer perceptron named NN as the other baseline. The number of hidden layers is 3, the number of neurons in each layer is set by searching [50, 100, 200, 400] and the learning rate is set to 0.1.

TCA & KPCA. Due to the limitation of time and memory, we randomly select 5000 samples from 14 subjects as source domain data while the target domain data remain the same. The kernel function is linear kernel and the regularization parameter μ is set to 1.

TPT. Multiple classifiers are learned on each source domain data using linear Support Vector Machine (LSVM), and the regularization parameter is set to 0.1. Multioutput Support Vector Regression (M-SVR) framework is utilized to learn the final mapping function $f(\cdot)$.

DAN. The feature extractor has two hidden layers with 100 neurons in each layer. The label predictor has one hidden layer with 100 neurons and a output layer. The domain classifier has one hidden layer with 100 neurons, one gradient reversal layer and one output layer. The learning rate is set to 0.1 and the parameter λ in gradient layer is set to 1.

IV. EXPERIMENT RESULTS

In this part, we will discuss experiment results of six different methods on SEED. We adopt leave-one-subject-out evaluation scheme in each case.

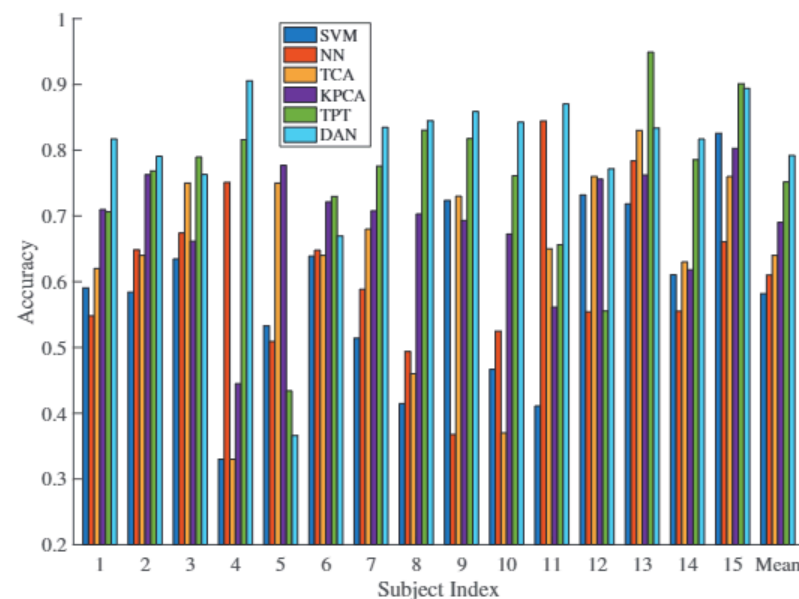


Figure 2. The accuracies of six methods (SVM, NN, TCA, KPCA, TPT, DAN) for each subject and the mean accuracies.

Table I
MEAN ACCURACIES AND STANDARD DEVIATIONS OF THE SIX
DIFFERENT METHODS

Methods	SVM	NN	TCA	KPCA	TPT	DAN
Mean	0.5818	0.6101	0.6400	0.6902	0.7517	0.7919
Std.	0.1385	0.1238	0.1466	0.0925	0.1283	0.1314

Figure 2 shows the accuracies of six different methods (SVM, NN, TCA, KPCA, TPT and DAN) for 15 subjects and Table I presents mean accuracies and standard deviations in detail. SVM has the lowest accuracy of 58.18%. Because the EEG signals vary between subjects, individual classifiers learned by SVM in source domain cannot distinguish emotion states in target domain precisely. By comparison, TCA and KPCA extract latent common components from both source and target data with low domain variance. These extracted features can improve the performance of classification. TPT achieves accuracy of 75.17%. It learns a mapping function that can transfer the classifiers trained in source domain to target domain instead of looking for a latent feature space between source and target data. Our DAN outperforms these method with the mean accuracy of **79.19%**. The standard deviation of DAN is not the smallest, because the result of subject 5 is extremely low.

To illustrate the transferability and discriminativeness of DAN learned features, we visualize the output of the second hidden layer from DAN and NN by projecting features into a 3-dimensional space using t-SNE [14] as shows in Figure 3. We then make the following observations: (1) NN features from different subjects are scattered in three emotion states, while DAN features are aligned much better between source and target domains. (2) With NN features, the datapoints are not discriminative enough, while with DAN features, all of the three emotion states can be evidently distinguished. These observations explain the superior performance of DAN over NN: (1) indicates that DAN can reduce the differences between source and target domain to learn domain-invariant features, which makes source classifiers more suitable for target data. (2)

进行三次间隔至少一周的实验，并要求在观看完电影片段后评估其真实的情绪反应。仅使用具有明确对应情绪的数据进行分析。关于 SEED 的更多细节，我们建议读者参考文献[2]。

B. 信号预处理与特征提取

对于信号预处理，由于原始脑电图（EEG）信号常受到肌电图（EMG）信号和眼电图（EOG）信号的干扰[12]，因此使用 1 Hz 到 75 Hz 之间的带通滤波器来滤除噪声和伪影。为了减小数据量，EEG 信号进一步下采样至 200 Hz。对于特征提取，从原始数据中提取微分熵（DE）特征，因为 DE 比功率谱密度（PSD）表现更优[13]。DE 特征包含五个频段：delta（1-4 Hz）、theta（4-8 Hz）、alpha（8-14 Hz）、beta（14-31 Hz）和 gamma（31-50 Hz）。每个频段包含 62 个 EEG 通道的特征，因此提取的 EEG 特征总维度为 310。

C. 参数详情

所有评估均采用留一受试者交叉验证进行。每次选择 15 个受试者中的 1 个作为目标域，其余 14 个作为源域。分类准确度在目标域上进行评估。

基线方法。我们采用支持向量机（SVM）作为我们的基线之一。我们使用线性核，并从-10 到 10 以 1 为步长搜索参数 C。为了与 DAN 进行比较，我们还采用传统的多层感知器 NN 作为另一个基线。隐藏层数为 3，每层的神经元数量通过搜索[50, 100, 200, 400]设置，学习率设置为 0.1。

TCA & KPCA。由于时间和内存的限制，我们从 14 个受试者中随机选择 5000 个样本作为源域数据，而目标域数据保持不变。核函数为线性核，正则化参数 μ 设置为 1。

TPT。使用线性支持向量机（LSVM）在每个源域数据上学习多个分类器，正则化参数设置为 0.1。采用多输出支持向量回归（M-SVR）框架来学习最终的映射函数 $f(\cdot)$ 。

DAN。特征提取器有两个隐藏层，每层有 100 个神经元。标签预测器有一个隐藏层，包含 100 个神经元和一个输出层。领域分类器有一个隐藏层，包含 100 个神经元、一个梯度反转层和一个输出层。学习率设置为 0.1，梯度层中的参数 λ 设置为 1。

IV. E R 在这一部分，我们将讨论在 SEED 上六种不同方法的实验结果。每种情况下我们都采用留一被试外评估方案。

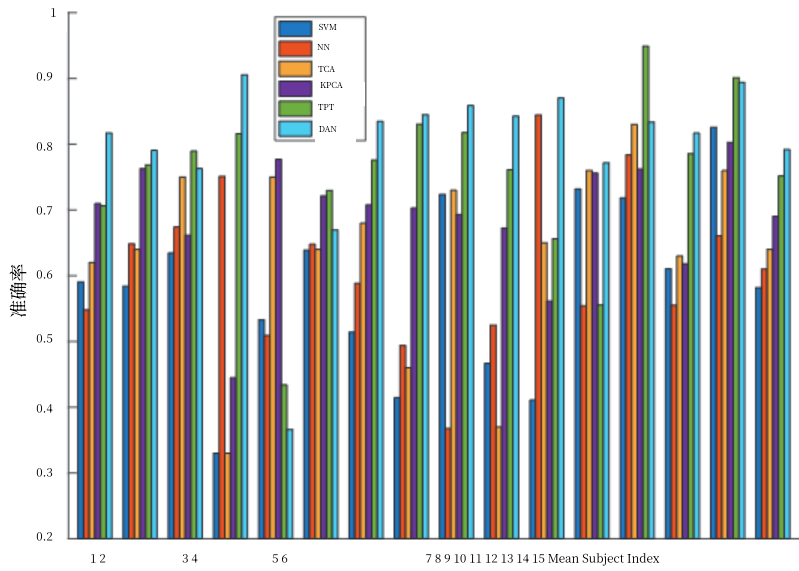


图 2. 六种方法（SVM、NN、TCA、KPCA、TPT、DAN）每个被试的准确率和平均准确率。

表 I									
M EAN ACCURACIES AND STANDARD DEVIATIONS OF THE SIX									
DIFFERENT METHODS									
方法	SVM	NN	TCA	KPCA	TPT	DAN	平均	标准差	
	0.5818	0.7517	0.7919	0.1385	0.1238	0.1466	0.5818	0.6101	0.6400
	0.6902	0.1314	0.0925	0.1283	0.1314	0.1314	0.6902	0.1314	0.1314

图 2 显示了六种不同方法（SVM、NN、TCA、KPCA、TPT 和 DAN）对 15 个受试者的准确率，表 1 详细列出了平均准确率和标准差。SVM 的准确率最低，为 58.18%。由于脑电信号在受试者之间存在差异，SVM 在源域中学习的分类器无法精确区分目标域中的情绪状态。相比之下，TCA 和 KPCA 从源域和目标数据中提取具有低域差异的潜在共同成分。这些提取的特征可以提高分类性能。TPT 达到了 75.17% 的准确率。它学习了一个映射函数，可以将源域中训练的分类器迁移到目标域，而不是寻找源域和目标数据之间的潜在特征空间。我们的 DAN 在这些方法中表现最佳，平均准确率为 79.19%。DAN 的标准差不是最小的，因为受试者 5 的结果极低。

为了说明 DAN 学习特征的迁移性和区分性，我们通过 t-SNE [14] 将 DAN 和 NN 的第二隐藏层输出特征投影到三维空间中进行可视化，如图 3 所示。然后我们得出以下观察结果：(1) NN 特征在不同被试者中分散在三种情绪状态中，而 DAN 特征在源域和目标域之间对齐得更好。(2) 使用 NN 特征时，数据点区分性不足，而使用 DAN 特征时，三种情绪状态可以明显区分。这些观察结果解释了 DAN 相对于 NN 的优越性能：(1) 表明 DAN 可以减少源域和目标域之间的差异，学习领域不变特征，这使得源分类器更适合目标数据。(2)

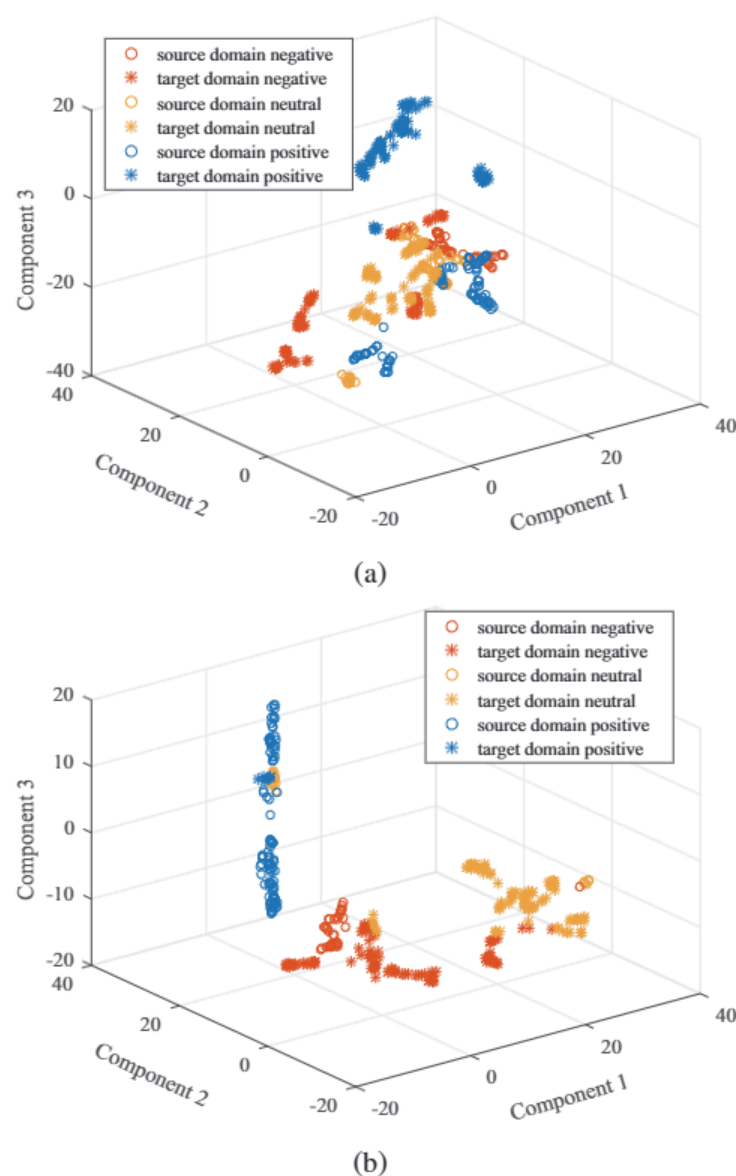


Figure 3. The effect of domain adaptation on the distributions of extracted features. Figures (a) and (b) denote NN and DAN, respectively, which both present all three different emotion states together from different domains.

implies that different emotion states are more easily identified with DAN features, which guarantees the accuracy of label predictor.

V. CONCLUSION AND FUTURE WORK

In this paper, we have proposed a neural network architecture for EEG-based emotion recognition with transfer learning techniques. Domain-invariant features are learned by backpropagation after adding a gradient reversal layer to domain classifier. We have compared the performance with five different methods: SVM, NN, TCA, KPCA, and TPT for SEED. The experimental results show that domain adaptation network (DAN) exceeds the other approaches in terms of accuracy, and achieves a 21.01% increase compared with SVM and a 18.18% increase compared with NN. We also search different parameters for DAN to find the optimal values. Our future work will focus on applying our proposed model to more categories of emotions, as well as to domain adaptation on subjects from different culture backgrounds.

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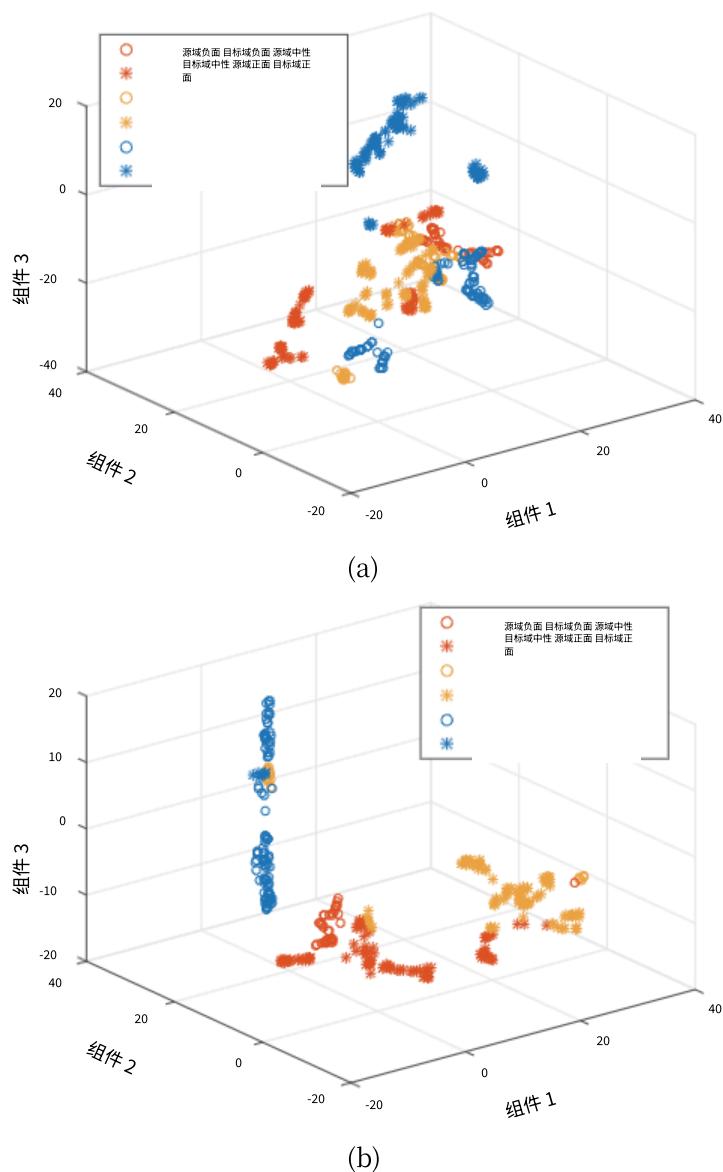


图 3. 域适应对提取特征分布的影响。图(a)和图(b)分别表示 NN 和 DAN，两者都展示了来自不同域的三个不同情绪状态。

这意味着使用 DAN 特征更容易识别不同的情绪状态，这保证了标签预测器的准确性。

V. 结论与未来工作

在本文中，我们提出了一种基于脑电图的情绪识别神经网络架构，并应用了迁移学习技术。通过在域分类器中添加一个梯度反转层，通过反向传播学习域不变特征。我们将性能与五种不同的方法进行了比较：SVM、NN、TCA、KPCA 和 TPT 用于 SEED。实验结果表明，域适应网络（DAN）在准确性方面优于其他方法，与 SVM 相比提高了 21.01%，与 NN 相比提高了 18.18%。我们还搜索了 DAN 的不同参数以找到最佳值。我们的未来工作将集中于将我们提出的模型应用于更多类别的情绪，以及在不同文化背景的受试者上进行域适应。

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